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University of Washington, Seattle WA Sep 2020 - Jun 2024
B.S. in Computer Science, B.A. in Mathematics
 Cumulative GPA: 3.98/4.00
 Honors: Magna Cum Laude, Phi Beta Kappa, Annual Dean's List

- Under review at *American Control Conference 2026*.

- Under revision at *Involve Undergraduate Mathematics Journal* 2025.

- Accepted to *Algorithms for Molecular Biology* 2024.

- Accepted to *Robotics: Science and Systems 2023*, awarded Best System Paper Finalist.

- Accepted to *ACM SIGACCESS Conference on Computers and Accessibility 2022*.

- Exploring various applications of stochastic processes to sketching/streaming algorithms.
- Designing a new super heavy hitter algorithm that uses sublogarithmic bits via Morris counters, a step towards solving ℓ_2 heavy hitters in optimal $O(1/\epsilon^2 \log(n))$ space.
- Studied state-of-the-art algorithms such as BPTree (Braverman et al.) and Velusamy-Yu.

- Independent reading on the KLS conjecture, including Bo'az Klartag's recent $O(\sqrt{\log n})$ bound on the isoperimetric constant as well as related works by Eldan and Lee-Vempala.
- Produced write-ups deriving the exponential-tilt formulation of Eldan's stochastic localization process and the logarithmic time horizon necessary to bound the process covariance.

- Designed exponential-time approximation algorithms for various CSPs including MAX- k LIN, inspired by the $7/8 + \varepsilon$ approximation algorithm for MAX-3SAT that runs in $2^{O(\varepsilon n)}$.

- Single-authored a paper ([arXiv](#)) introducing a new optimal control algorithm that extends lossless convexification theory from stepwise to piecewise-linear parameterized controls.
- Proved algorithm correctness using tools from convex analysis and optimization theory.

	Washington Experimental Mathematics Lab , UW, Seattle WA Sep 2024 - Dec 2024 <i>Advisor: Stefan Steinerberger</i> - Co-authored a paper (arXiv) examining a class of dynamical systems in the complex plane. - Discovered and proved results for the stochastic variant, including our main theorem regarding the existence and expectation of the stationary measure in higher dimensions. - Presented at the <i>Northwest Undergrad Math Symposium 2024</i>, receiving 2nd place prize.
	WEIRD Robot Learning Lab , UW, Seattle WA Jan 2023 - Jun 2023 <i>Advisor: Abhishek Gupta</i> Collaborated on a paper (arXiv) introducing generative models as an effective data augmentation technique for robot learning, winning Best System Paper Finalist at <i>RSS 2023</i>.
	Matsen Group , Fred Hutchinson Cancer Center, Seattle WA Jun 2021 - Sep 2021 <i>Advisor: Frederick Matsen</i> - Contributed to a paper (arXiv) introducing the sDAG: a data structure that compactly represents a collection of phylogenetic trees in order to perform Bayesian inference. - Developed fast tree rearrangement algorithms necessary for MCMC steps.
	Makeability Lab , UW, Seattle WA Jun 2020 - Jun 2022 <i>Advisor: Jon Froehlich</i> - Started an initiative to use deep learning for automatic sidewalk accessibility evaluation. - Co-authored a paper (dl.acm.org) examining model performance across international cities.
WORK EXPERIENCE	Software Engineering Intern , Apple, San Diego CA Jun 2023 - Sep 2023 - Introduced an improved adaptive bitrate algorithm reducing runtime from $O(n^2)$ to $O(n)$. Software Engineering Intern , Microsoft, Redmond WA Jun 2022 - Sep 2022 - Designed a custom image segmentation algorithm reducing runtime from 80 ms to 0.5 ms.
TEACHING EXPERIENCE	MATH 54 - Linear Algebra , GSI, Berkeley CA Aug 2025 - Dec 2025 - Led 6 hours of weekly discussion sections on linear algebra and differential equations. - Wrote exams, graded assignments, and held office hours, earning a 6.5/7 course evaluation. Math Tutor , Fusion Math, Seattle WA Jan 2025 - Jun 2025 - Taught lessons covering curriculum in AP Calculus. CS Tutor , Self-employed, Seattle WA Mar 2022 - Jun 2022 - Taught lessons covering curriculum in AP Computer Science. Cello Teacher , Seattle Music Partners Sep 2016 - Jun 2020 - Taught cello lessons to students at underserved elementary schools.
PERSONAL PROJECTS	Euclideanate Nov 2024 - Present - A highly optimized algorithm for finding minimal-step straightedge-and-compass constructions, using symmetry and field-degree constraints to prune the search space (GitHub). DanceTime May 2021 - Jul 2023 - A <i>Just Dance</i> remake with a custom gesture scoring algorithm using pose estimation, conditional filtering, polynomial regression, and numerical optimization (GitHub). AlphaFour Jun 2022 - Jul 2023 - A <i>Connect 4</i> AI that learns via self-play deep reinforcement learning (GitHub). GeoKnowr Nov 2022 - Dec 2022 - A <i>GeoGuessr</i> AI that reliably guesses within 2000 km of the ground truth (GitHub).
EXTRA-CURRICULARS	UW Formula Motorsports , Engineer Jan 2023 - Jun 2024 Designed the team's first driverless steering system (motor + custom rack and pinion).